Jadavpur University

Department of Computer Science and Engineering

Assignment 6: Comprehensive Local & Wide Area Network Modeling in Cisco Packet Tracer

Point-to-Point Links, Shared Collision Domains, Layer-2 Switching & CAM/ARP Dynamics, Subnet Bridging, Multi-Router WAN Static Routing, Dynamic Host Configuration (DHCP), and Enterprise Multi-Tier Services (DNS, Web, FTP)

Author Information:
Name: Sampad De
Roll Number: 002410501025
Degree: B.E. Computer Science & Engineering
Course: CSE/PC/B/S/314 Computer Networks Lab
Simulation Engine: Cisco Packet Tracer v9.0.1
Host Environment: macOS (Apple Silicon / Darwin)

Laboratory Details:
Assignment: 6 (Packet Tracer Modeling)
Router Hardware: Cisco 1841 / 2911
Switch Hardware: Cisco Catalyst 2950-24 / 2960
Submission Date: November 2026

Table of Contents

1 Simulation Environment & Device Taxonomy . . . 2

2 Question 1: Direct Host-to-Host Crossover Link . . . 3

2.1 Problem Statement & Physical Layer Operation . . . 3

2.2 Configuration & Verification . . . 3

3 Question 2: Hub-Based Local Area Network (LAN-A) . . . 4

3.1 Problem Statement & Mechanism of Operation . . . 4

3.2 Addressing Scheme . . . 4

4 Question 3: Switch-Based LAN (LAN-B) & CAM/ARP Dynamics . . . 8

4.1 Problem Statement & Theoretical Framework . . . 8

4.2 Pre-Communication Baseline . . . 8

4.3 Traffic Generation & Post-Communication State Evolution . . . 10

5 Question 4: Inter-LAN Bridging (Hub to Switch Interconnection) . . . 11

5.1 Problem Statement & Subnet Boundary Principle . . . 11

5.2 Uplink Port Multi-MAC Address Learning . . . 13

6 Question 5: Multi-Router Inter-Network Routing over Serial WAN . . . 14

6.1 Architecture & Interface Configuration . . . 14

7 Question 6: Dynamic Host Configuration Protocol (DHCP) . . . 17

7.1 Problem Statement & DORA Protocol Exchange . . . 17

8 Question 7: Enterprise Multi-Tier Services (DHCP, DNS, Web & FTP) . . . 19

8.1 Architecture & Infrastructure Provisioning . . . 19

8.2 Client Dynamic Leasing & Application Verifications . . . 20

9 Key Takeaways & Conclusion . . . 22

10 References . . . 22

1 Simulation Environment & Device Taxonomy

Cisco Packet Tracer is a discrete-event network modeling environment that simulates protocol behavior across all OSI layers. Table 1 summarizes the structural and operational differences between the devices deployed in this assignment.

Table 1: Operational comparison of networking devices deployed in Assignment 6.

Device Type	OSI Layer	Forwarding Logic	Collision Domains	Broadcast Domain
Hub	Layer 1	Electrical Repeat	1 (Shared across all ports)	1 (All connected nodes)
Switch	Layer 2	Hardware Frame Switching	N (1 per physical port)	1 (Flat VLAN domain)
Router	Layer 3	Routing Table Lookup (RIB)	N (1 per physical port)	N (1 per interface)
DHCP Server	Layer 7	Dynamic IP/Gateway Lease	N/A (Application Layer)	Bounded by L3 Router
DNS Server	Layer 7	FQDN-to-IP Translation	N/A (Application Layer)	Bounded by L3 Router
Web/FTP Server	Layer 7	Application Services	N/A (Application Layer)	Bounded by L3 Router

2 Question 1: Direct Host-to-Host Crossover Link

2.1 Problem Statement & Physical Layer Operation

Connect two PC endpoints directly using a Copper Cross-Over cable, assign static IP addresses in the same subnet, and verify connectivity using ICMP ping.

In twisted-pair Ethernet (10BASE-T / 100BASE-TX), PC NICs operate as Medium Dependent Interface (MDI) devices, transmitting on pins 1 and 2 ($TX+, TX-$) and receiving on pins 3 and 6 ($RX+, RX-$). Connecting two MDI ports with a straight-through cable would short transmit pins to transmit pins. A crossover cable swaps pins 1 and 2 with pins 3 and 6, coupling the transmit pins of one host directly to the receive pins of the other.

2.2 Configuration & Verification

- **PC0:** IP `192.168.2.1`, Subnet Mask `255.255.255.0` (/24)
- **PC1:** IP `192.168.2.4`, Subnet Mask `255.255.255.0` (/24)
- **Cable:** Copper Cross-Over

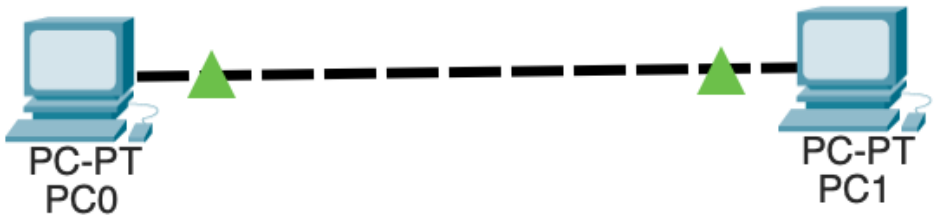


Figure 1: Question 1: Direct back-to-back host topology connected with a Copper Cross-Over cable.

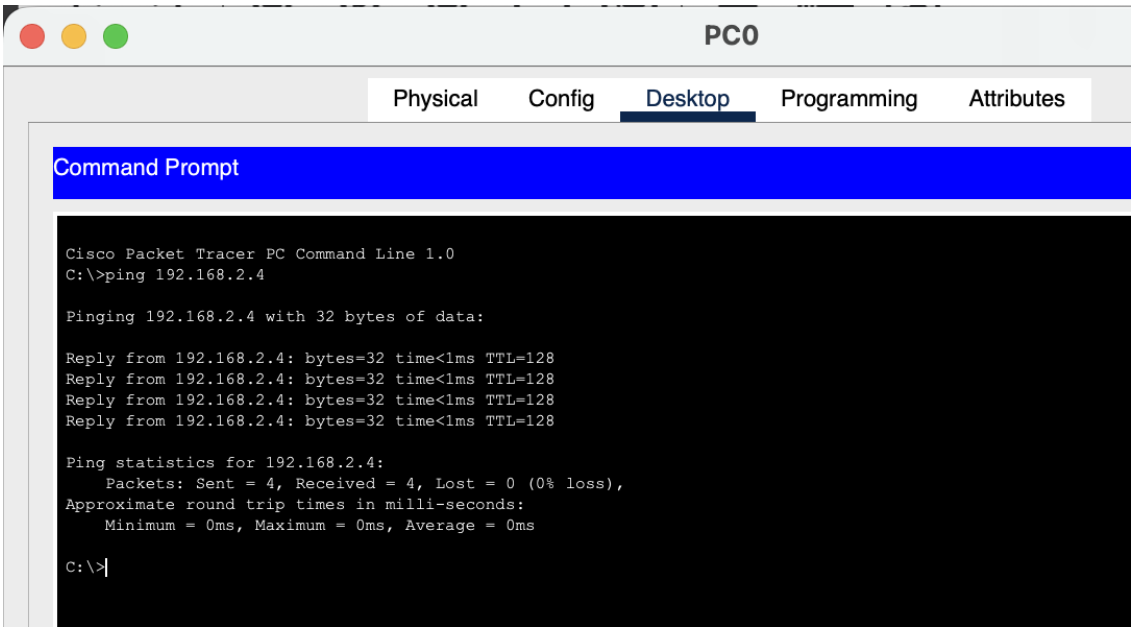


Figure 2: Question 1: ICMP ping execution on PC0 targeting PC1 (192.168.2.4) showing 0% loss and < 1 ms RTT.

Inference: Direct communication between two computing systems is achieved without intermediate hubs, switches, or routers when physical pinouts match and hosts reside on the same IP subnet.

3 Question 2: Hub-Based Local Area Network (LAN-A)

3.1 Problem Statement & Mechanism of Operation

Construct a 3-host local area network named **LAN-A** using a central hub (`Hub-PT`) and verify pairwise connectivity between all hosts.

A network Hub operates strictly at Layer 1. When an electrical signal enters any port, the repeater regenerates it onto all other ports:

- **Single Collision Domain:** All three hosts share one physical transmission channel. Simultaneous transmissions cause physical collisions requiring CSMA/CD backoff.
- **Half-Duplex:** Hosts cannot transmit and receive simultaneously.
- **Shared Medium:** Every packet is delivered to all connected stations, regardless of destination MAC.

3.2 Addressing Scheme

- `LAN-A-PC0`: `192.168.2.1` / `255.255.255.0`
- `LAN-A-PC1`: `192.168.2.2` / `255.255.255.0`
- `LAN-A-PC2`: `192.168.2.3` / `255.255.255.0`

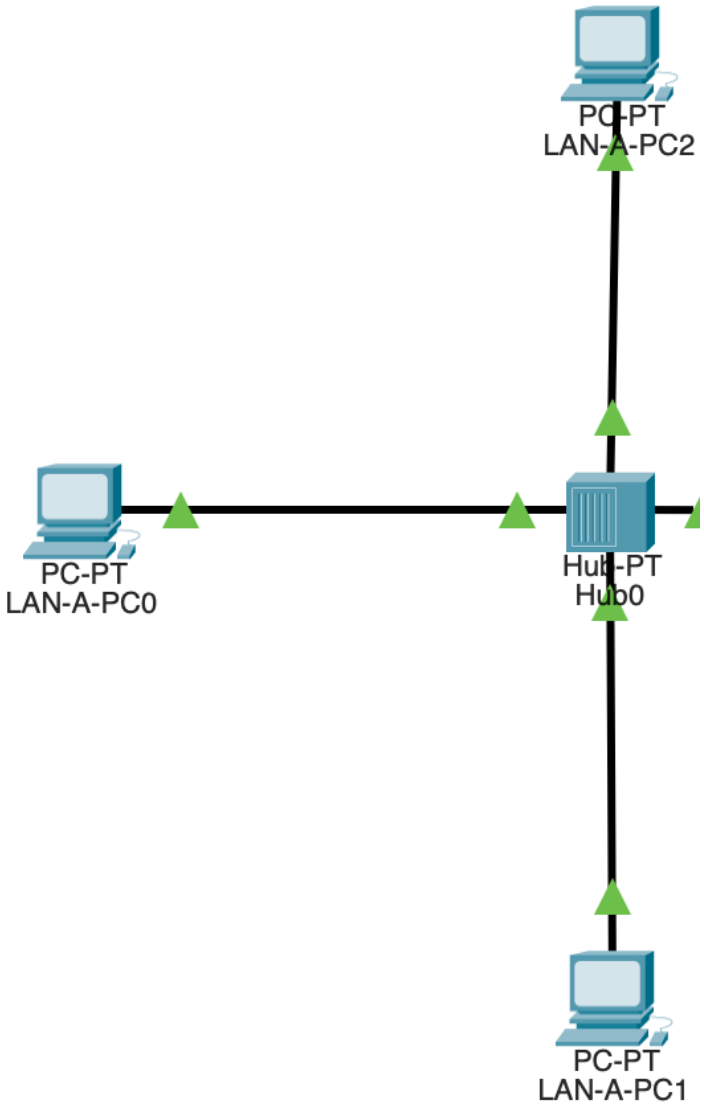


Figure 3: Question 2: LAN-A hub-based star topology with 3 workstations.

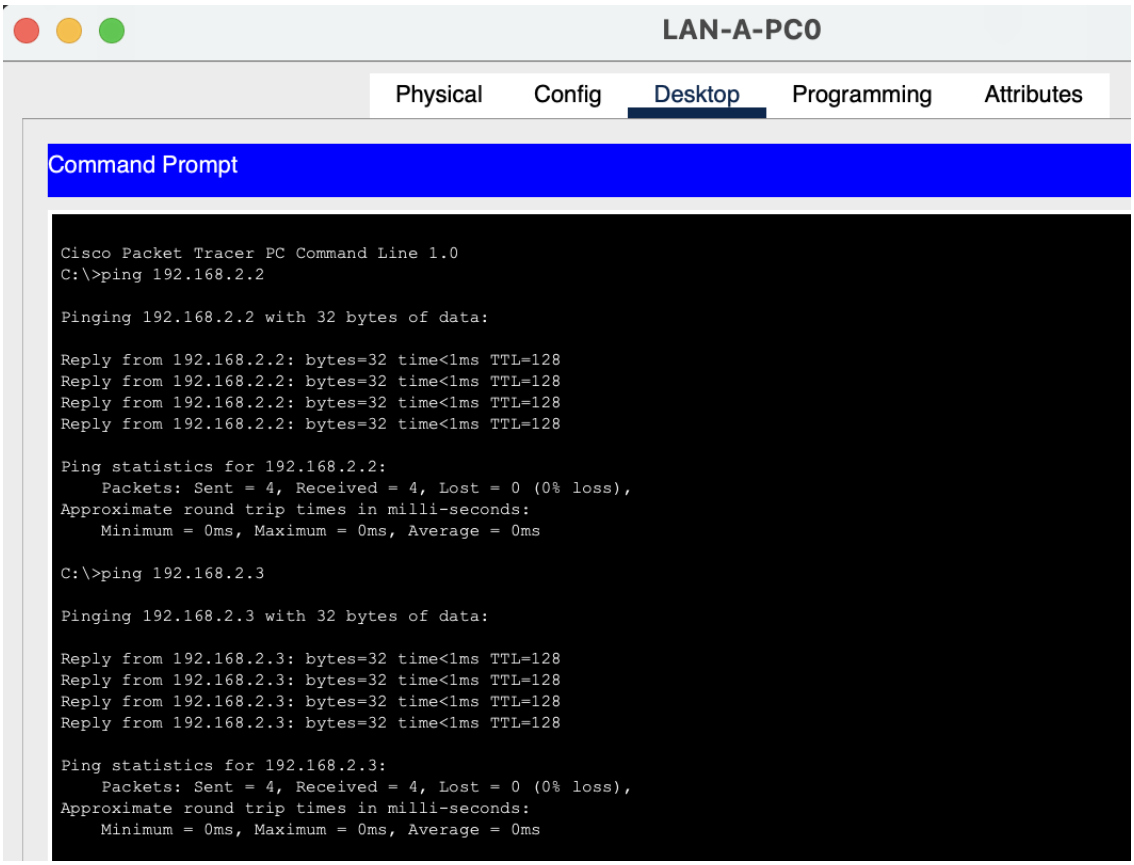


Figure 4: Question 2: Pairwise ping execution on PC0 targeting PC1 (192.168.2.2) and PC2 (192.168.2.3).

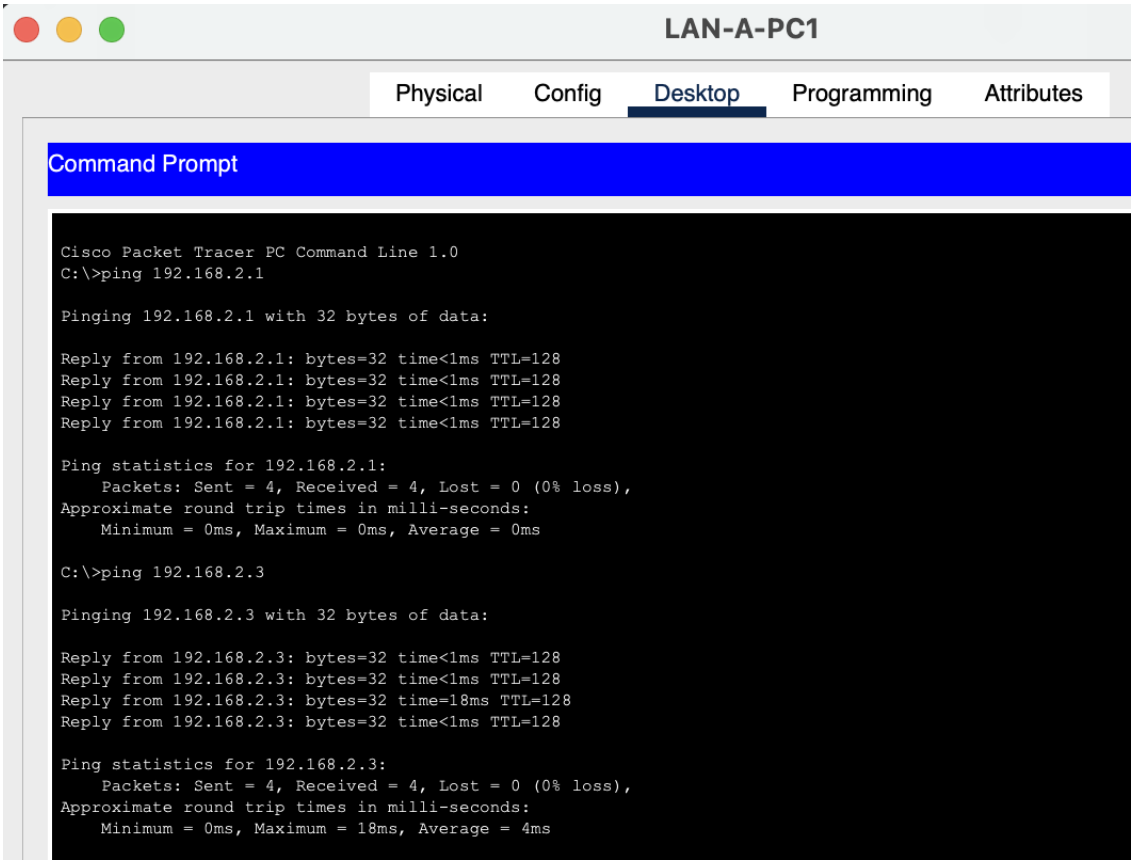


Figure 5: Question 2: Pairwise ping execution on PC1 targeting PC0 (192.168.2.1) and PC2 (192.168.2.3).

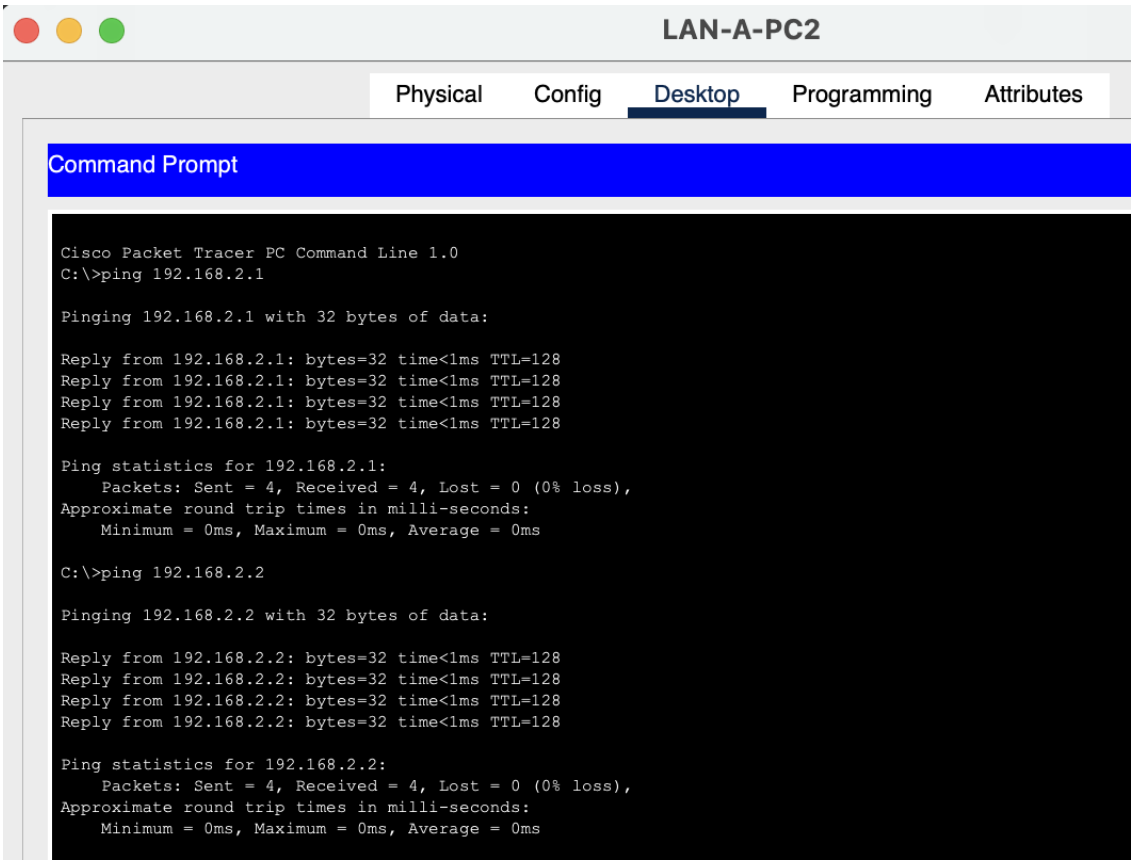


Figure 6: Question 2: Pairwise ping execution on PC2 targeting PC0 (192.168.2.1) and PC1 (192.168.2.2).

Inference: All pairwise communications succeeded with 0% packet loss. However, every transmission

occupied the shared bandwidth of all connected links simultaneously.

4 Question 3: Switch-Based LAN (LAN-B) & CAM/ARP Dynamics

4.1 Problem Statement & Theoretical Framework

Construct **LAN-B** using a Cisco Catalyst 2960 Layer-2 switch. Record host **ARP Tables** and the switch **MAC Address Table (CAM Table)** both *before* and *after* generating traffic.

- **Micro-segmentation:** Each switch port forms an isolated collision domain, operating in full-duplex.
- **CAM Learning Algorithm:** The switch inspects the source MAC of incoming frames and maps it to the ingress port. If the destination MAC is known, it forwards unicast frames directly to the mapped port; otherwise, it floods the frame.
- **ARP (RFC 826):** Hosts resolve 32-bit IPv4 addresses to 48-bit MAC addresses via broadcast requests and unicast replies.

4.2 Pre-Communication Baseline

Prior to generating any network traffic, the tables were inspected:

- `arp -a` on `LAN-B-PC1`: `No ARP Entries Found`.
- `show mac address-table` on the switch: Empty dynamic table.

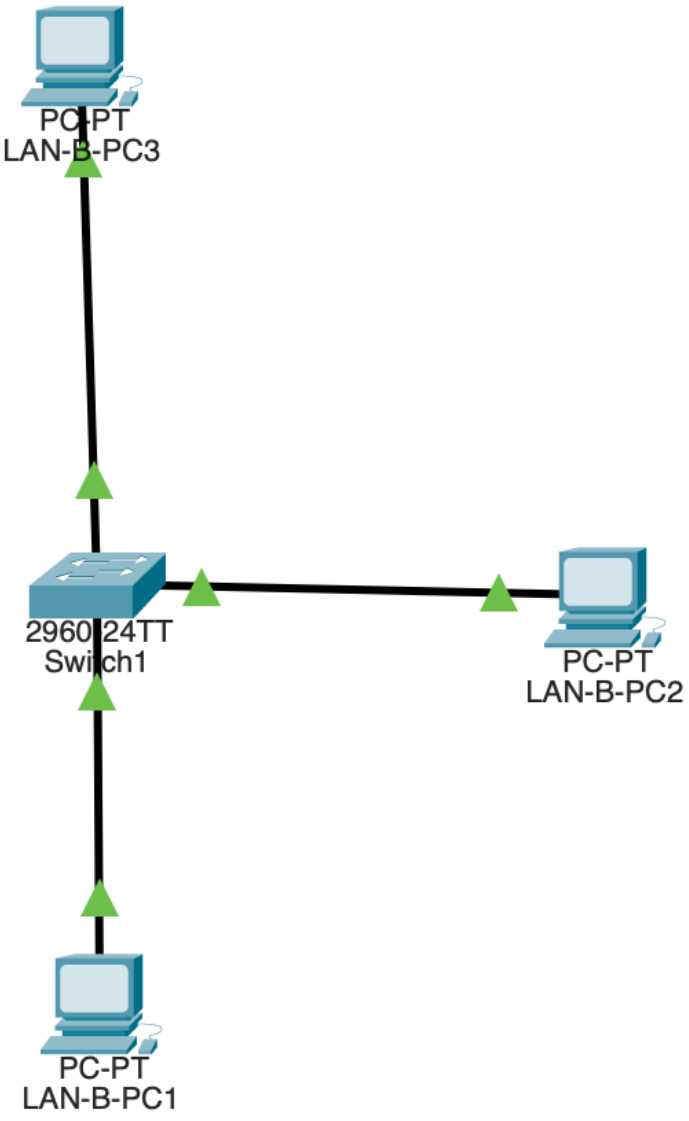
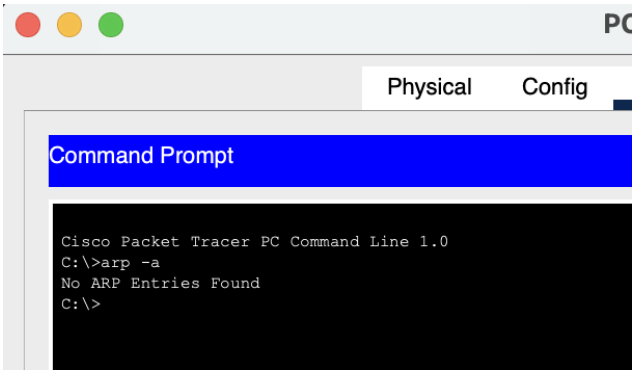


Figure 7: Question 3: LAN-B switch topology using a Cisco Catalyst 2960 switch.



(a) Pre-ping host ARP cache: empty.

```
Switch>enable
Switch#show mac-address table
                        ^
% Invalid input detected at '^' marker.

Switch#show mac address-table
          Mac Address Table
-------------------------------------------

Vlan    Mac Address       Type        Ports
----    -----------       --------    -----

Switch#enable
Switch#show mac address-table
          Mac Address Table
-------------------------------------------

Vlan    Mac Address       Type        Ports
----    -----------       --------    -----

Switch#
```

(b) Pre-ping switch CAM table: empty.

Figure 8: Question 3: Pre-communication baseline showing unpopulated state tables.

4.3 Traffic Generation & Post-Communication State Evolution

Workstation `LAN-B-PC1` executed pings targeting `PC2` (`192.168.2.2`) and `PC3` (`192.168.2.3`).

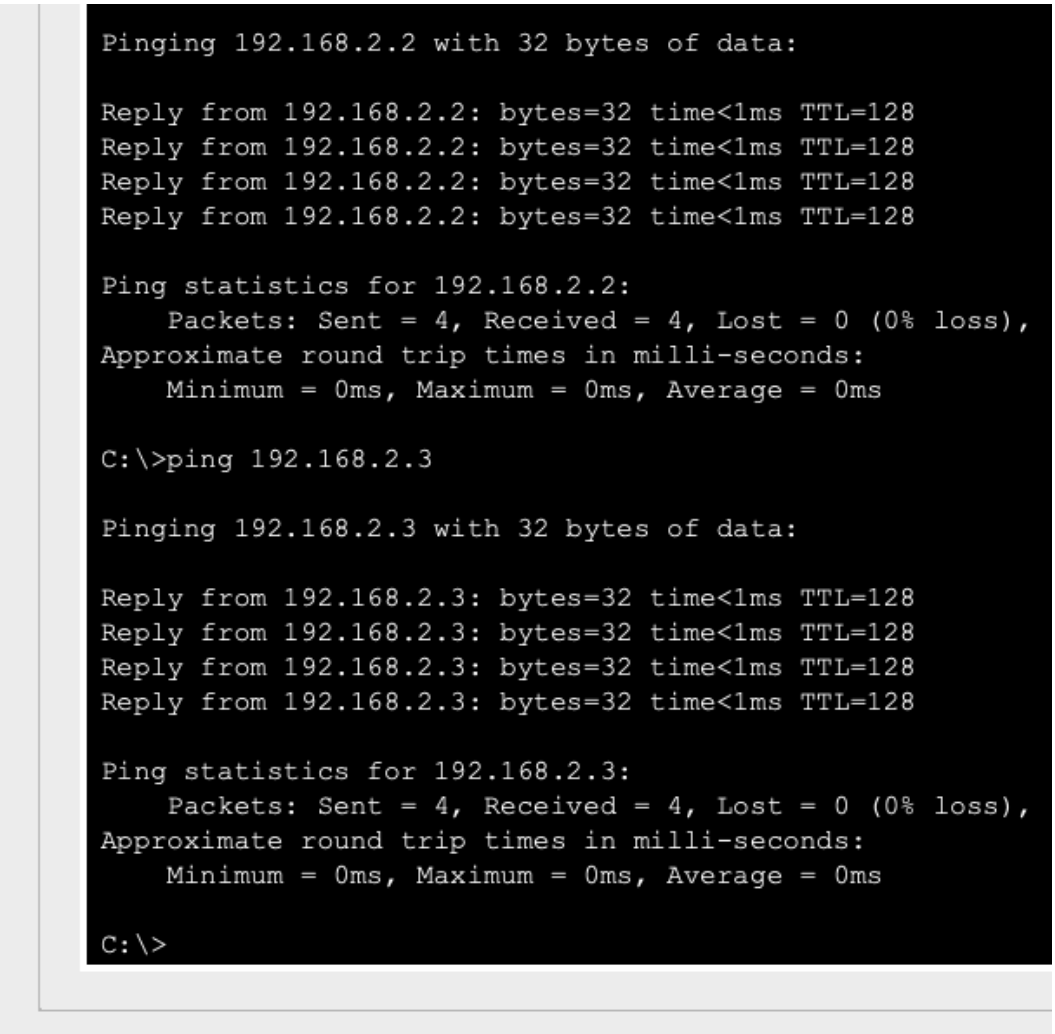
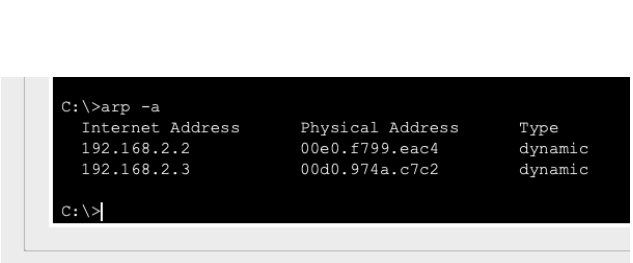
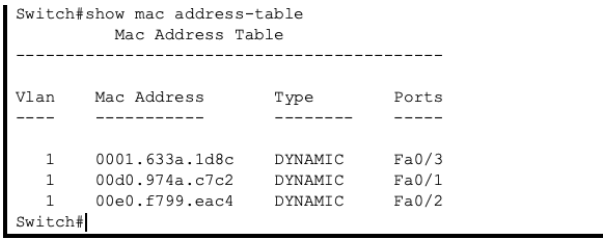


Figure 9: Question 3: ICMP ping sequence from LAN-B-PC1 to PC2 (192.168.2.2) and PC3 (192.168.2.3).



(a) Post-ping host ARP cache populated dynamically.



(b) Post-ping switch CAM table showing learned ports.

Figure 10: Question 3: Post-communication tables showing dynamic ARP and CAM resolution.

Inference:

1. `PC1`'s ARP table dynamically resolved `192.168.2.2` → `00e0.f799.eac4` and `192.168.2.3` → `00d0.974a.c7c2`.
2. The switch CAM table dynamically learned ports `Fa0/1`, `Fa0/2`, and `Fa0/3` for the respective MAC addresses, eliminating subsequent unicast flooding.

5 Question 4: Inter-LAN Bridging (Hub to Switch Interconnection)

5.1 Problem Statement & Subnet Boundary Principle

Connect LAN-A (Hub) and LAN-B (Switch) using a Copper Cross-Over cable, test connectivity across LANs, and observe ARP and switch CAM table changes.

Subnet Boundary Principle: A hub and a switch operate at Layer 1 and Layer 2. If LAN-A and LAN-B belonged to different IP subnets (e.g. `192.168.2.0/24` and `192.168.3.0/24`), hosts would drop frames because they look for an absent Layer-3 default gateway. To enable flat Layer-2 bridging across the crossover cable without a router, all six workstations were assigned IP addresses within the unified subnet `192.168.2.0/24`.

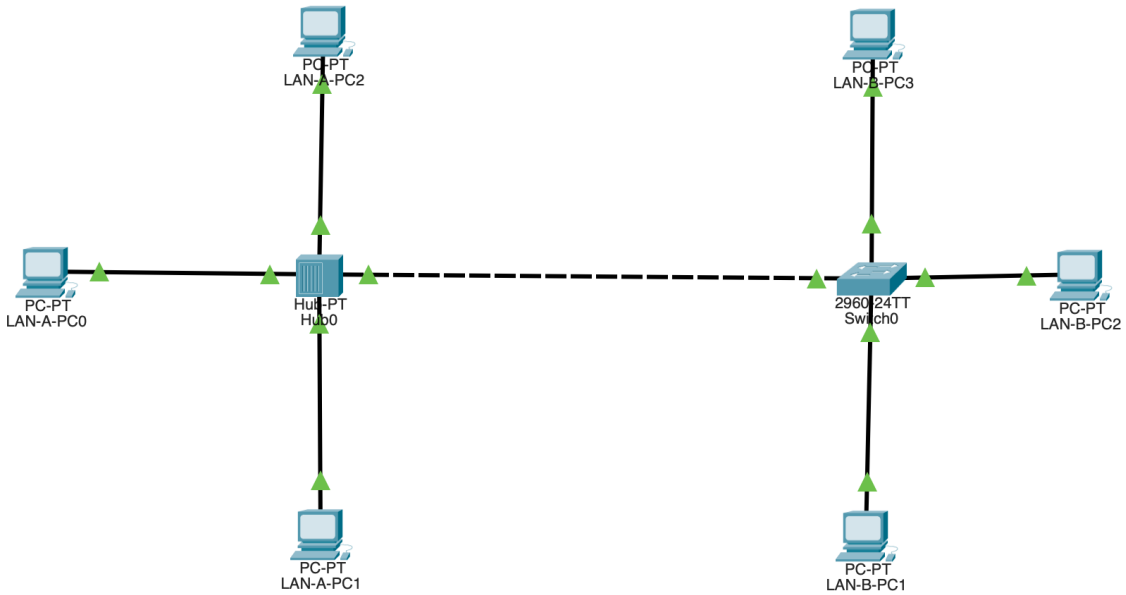


Figure 11: Question 4: Interconnected topology linking Hub0 to Switch0 via a crossover cable.

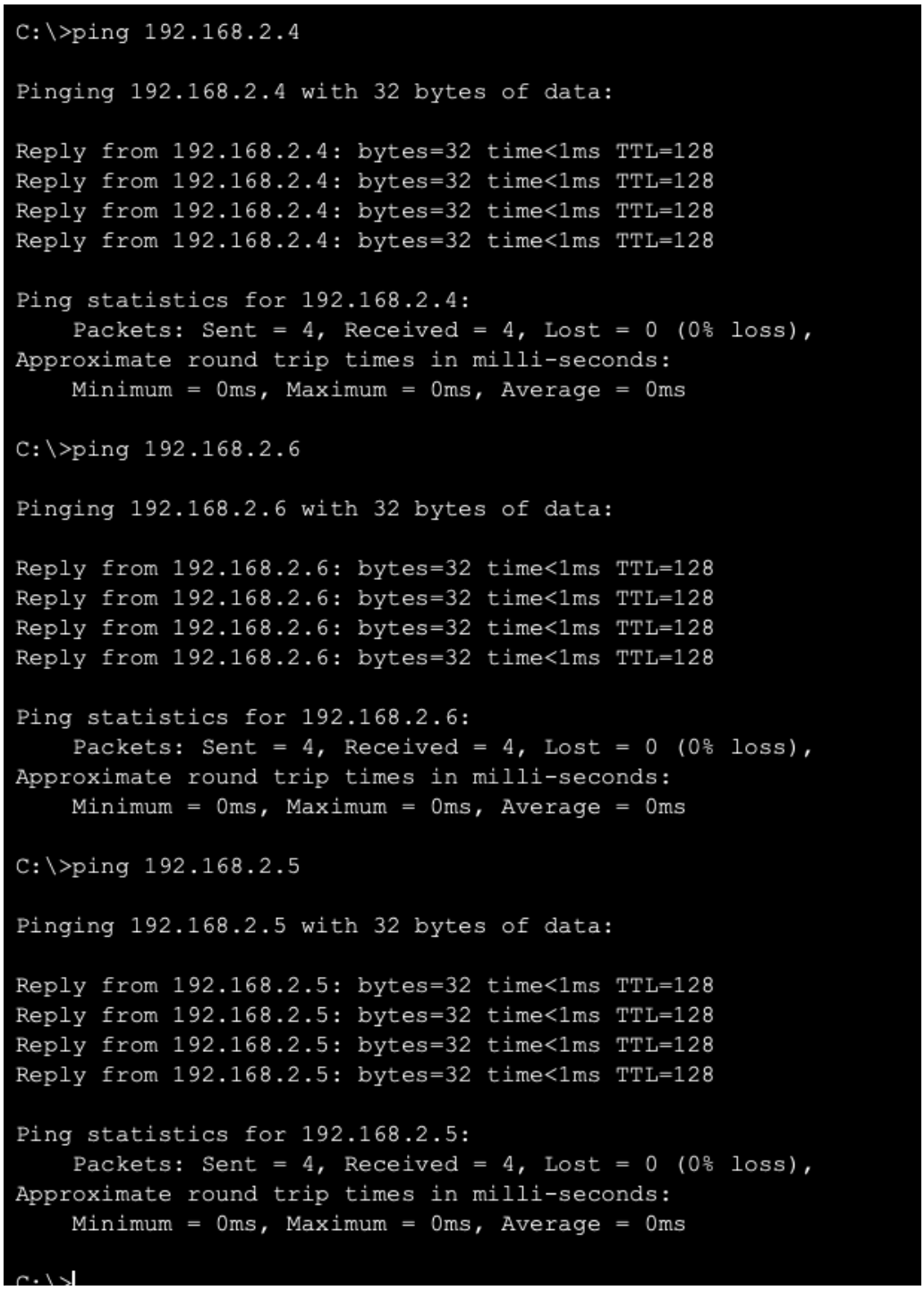
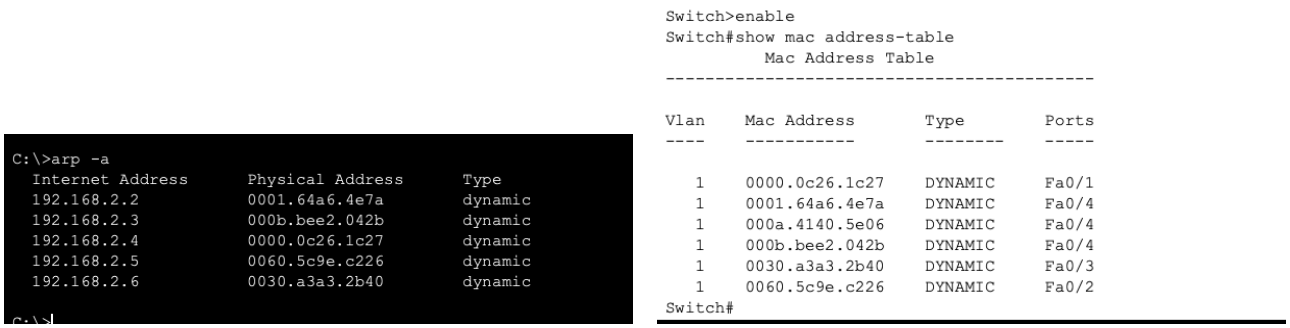


Figure 12: Question 4: Cross-LAN ping execution from LAN-A-PC1 to LAN-B hosts (192.168.2.4, .2.6, .2.5).

5.2 Uplink Port Multi-MAC Address Learning

Figure 13 displays the post-ping ARP table of `LAN-A-PC1` and the switch CAM table.



```
C:\>arp -a
  Internet Address      Physical Address      Type
  192.168.2.2           0001.64a6.4e7a        dynamic
  192.168.2.3           000b.bee2.042b        dynamic
  192.168.2.4           0000.0c26.1c27        dynamic
  192.168.2.5           0060.5c9e.c226        dynamic
  192.168.2.6           0030.a3a3.2b40        dynamic

C:\>
```

(a) LAN-A-PC1 ARP table containing all 5 nodes.

```
Switch>enable
Switch#show mac address-table
          Mac Address Table
-------------------------------------------

Vlan    Mac Address       Type        Ports
----    -----------       --------    -----

   1    0000.0c26.1c27    DYNAMIC     Fa0/1
   1    0001.64a6.4e7a    DYNAMIC     Fa0/4
   1    000a.4140.5e06    DYNAMIC     Fa0/4
   1    000b.bee2.042b    DYNAMIC     Fa0/4
   1    0030.a3a3.2b40    DYNAMIC     Fa0/3
   1    0060.5c9e.c226    DYNAMIC     Fa0/2
Switch#
```

(b) Switch CAM table showing multi-MAC on Fa0/4.

Figure 13: Question 4: State tables showing multi-MAC learning on the hub uplink port.

Key Observation: On the switch, ports `Fa0/1`, `Fa0/2`, and `Fa0/3` map to one MAC address each. Port `Fa0/4` (the link to the hub) dynamically learned **three MAC addresses** (`0001.64a6.4e7a`, `000a.4140.5e06`, `000b.bee2.042b`). This demonstrates that an uplink port connected to a shared or bridged segment legitimately associates multiple MAC addresses with a single physical switch port.

6 Question 5: Multi-Router Inter-Network Routing over Serial WAN

6.1 Architecture & Interface Configuration

Experiment 5 interconnects two enterprise campus LANs across a synchronous serial WAN link:

- **JU-Main LAN:** `192.168.148.0/24`, Gateway `192.168.148.1` (`Gig0/0` on `JU-Main-Router`).
- **JU-SL LAN:** `192.168.149.0/24`, Gateway `192.168.149.1` (`Gig0/0` on `JU-SL-Router`).
- **WAN Serial Link:** `192.168.150.0/24` over `WIC-2T` interfaces. `JU-Main-Router` acts as DCE (`clock rate 64000`).

Listing 1: Static route configurations on the routers.

```
! On JU-Main-Router:
ip route 192.168.149.0 255.255.255.0 192.168.150.2

! On JU-SL-Router:
ip route 192.168.148.0 255.255.255.0 192.168.150.1
```

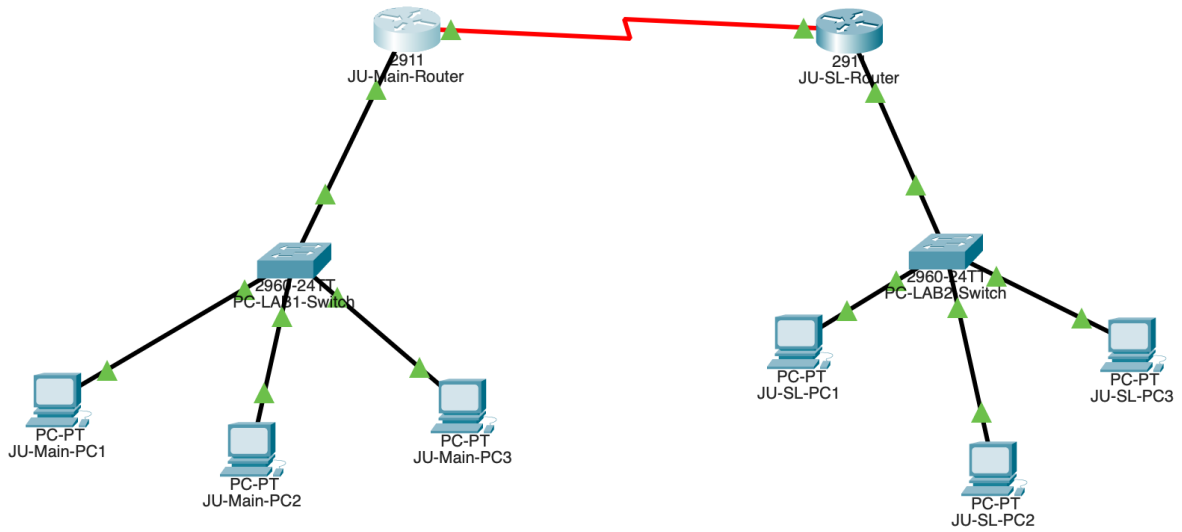


Figure 14: Question 5: Enterprise multi-router topology interconnecting JU-Main and JU-SL across a serial WAN.

```
JU-Main-Router>
JU-Main-Router>show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     192.168.148.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.148.0/24 is directly connected, GigabitEthernet0/0
L       192.168.148.1/32 is directly connected, GigabitEthernet0/0
S    192.168.149.0/24 [1/0] via 192.168.150.2
     192.168.150.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.150.0/24 is directly connected, Serial0/0/0
L       192.168.150.1/32 is directly connected, Serial0/0/0

JU-Main-Router>ping 192.168.150.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.150.2, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 13/20/39 ms

JU-Main-Router>
```

Figure 15: Question 5: Cisco IOS routing tables on JU-Main-Router and JU-SL-Router verifying static routes.

```
JU-Main-Router>ping 192.168.150.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.150.2, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 14/18/28 ms

JU-Main-Router>
```

Figure 16: Question 5: Point-to-point WAN ping test across serial link (100% success rate).

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.149.2

Pinging 192.168.149.2 with 32 bytes of data:

Request timed out.
Reply from 192.168.149.2: bytes=32 time=26ms TTL=126
Reply from 192.168.149.2: bytes=32 time=1ms TTL=126
Reply from 192.168.149.2: bytes=32 time=1ms TTL=126

Ping statistics for 192.168.149.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 26ms, Average = 9ms

C:\>ping 192.168.149.2

Pinging 192.168.149.2 with 32 bytes of data:

Reply from 192.168.149.2: bytes=32 time=18ms TTL=126
Reply from 192.168.149.2: bytes=32 time=29ms TTL=126
Reply from 192.168.149.2: bytes=32 time=19ms TTL=126
Reply from 192.168.149.2: bytes=32 time=25ms TTL=126

Ping statistics for 192.168.149.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 18ms, Maximum = 29ms, Average = 22ms
```

(a) JU-Main-PC1 pinging remote JU-SL-PC1.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.148.2

Pinging 192.168.148.2 with 32 bytes of data:

Reply from 192.168.148.2: bytes=32 time=11ms TTL=126
Reply from 192.168.148.2: bytes=32 time=20ms TTL=126
Reply from 192.168.148.2: bytes=32 time=23ms TTL=126
Reply from 192.168.148.2: bytes=32 time=22ms TTL=126

Ping statistics for 192.168.148.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 11ms, Maximum = 23ms, Average = 19ms
```

(b) JU-SL-PC1 pinging remote JU-Main-PC1.

Figure 17: Question 5: Bidirectional ping verification between workstations across the routed WAN.

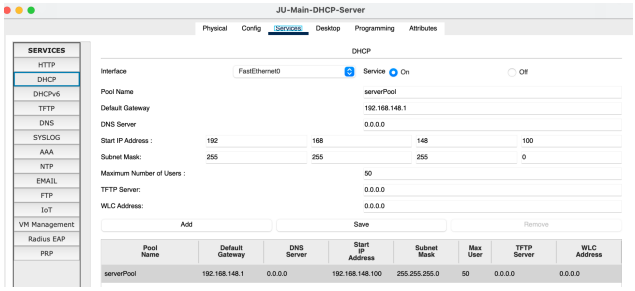
Analysis of Initial Packet Timeout: In Figure 17a, the first ping packet reported `Request timed out`. This is an expected artifact of a double ARP resolution: `JU-Main-PC1` had to resolve its default gateway, and `JU-SL-Router` had to resolve the destination host's MAC address. Once cached, subsequent pings achieved 100% success with TTL = 126 (128 − 2 router hops).

7 Question 6: Dynamic Host Configuration Protocol (DHCP)

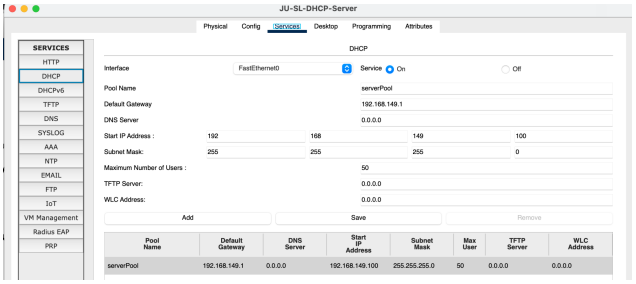
7.1 Problem Statement & DORA Protocol Exchange

Deploy dedicated DHCP servers in each LAN (`JU-Main` and `JU-SL`) and configure all workstations to obtain IP addresses and default gateways dynamically via the four-stage DORA exchange (Discover, Offer, Request, Acknowledge).

- `JU-Main-DHCP-Server`: Static IP `192.168.148.10`, Gateway `192.168.148.1`, Pool `192.168.148.100+`.
- `JU-SL-DHCP-Server`: Static IP `192.168.149.10`, Gateway `192.168.149.1`, Pool `192.168.149.100+`.

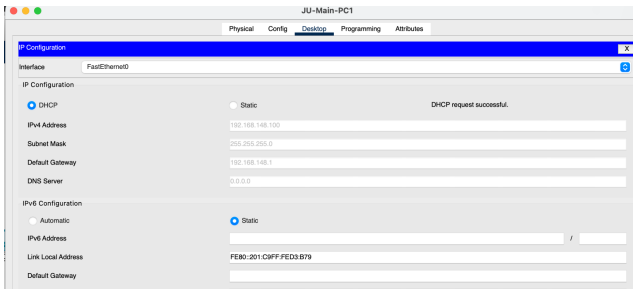


(a) JU-Main DHCP service configuration.

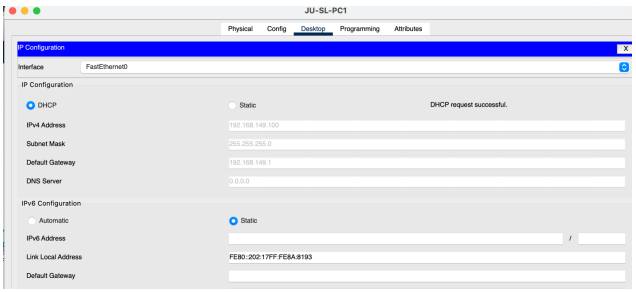


(b) JU-SL DHCP service configuration.

Figure 18: Question 6: DHCP service pool definitions with gateway and IP ranges.



(a) JU-Main-PC1 successful DHCP lease.



(b) JU-SL-PC1 successful DHCP lease.

Figure 19: Question 6: Workstation IP panels displaying "DHCP request successful!".



Figure 20: Question 6: End-to-end ping verification between DHCP-configured workstations across the routed WAN.

Inference: Dynamic leasing completed successfully, and end-to-end reachability was verified between hosts across the routed WAN at dynamically assigned IP `192.168.149.100`.

8 Question 7: Enterprise Multi-Tier Services (DHCP, DNS, Web & FTP)

8.1 Architecture & Infrastructure Provisioning

Experiment 7 deploys a complete departmental network (**CSE**) in domain `cse.myuniv.edu` on subnet `192.168.151.0/24`:

- **DHCP Server (‘192.168.151.2‘):** Dynamic IP leasing with DNS server option `192.168.151.3`.
- **DNS Server (‘192.168.151.3‘):** A-Records: `www.cse.myuniv.edu` → `192.168.151.4`, `ftp.cse.myuniv.edu` → `192.168.151.5`.
- **HTTP Web Server (‘192.168.151.4‘):** Departmental web page.
- **FTP Server (‘192.168.151.5‘):** Authenticated user `cisco` with full privileges.

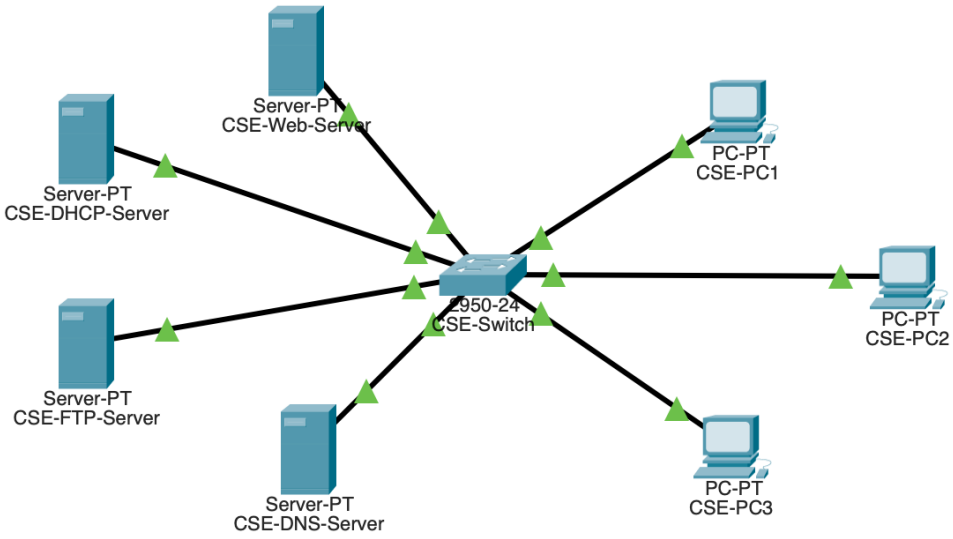
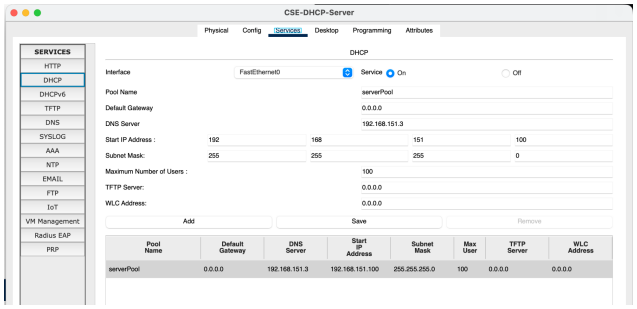
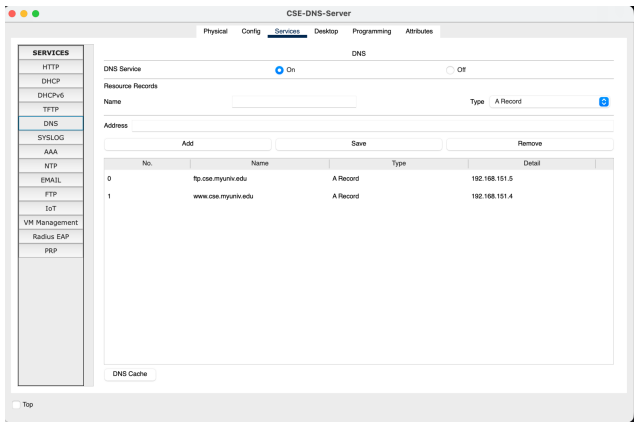


Figure 21: Question 7: Complete enterprise CSE LAN topology with 4 dedicated servers.



(a) DHCP service pool delivering DNS server pointer.



(b) Authoritative DNS A-records in cse.myuniv.edu.

Figure 22: Question 7: Core network service configurations (DHCP and DNS).

8.2 Client Dynamic Leasing & Application Verifications

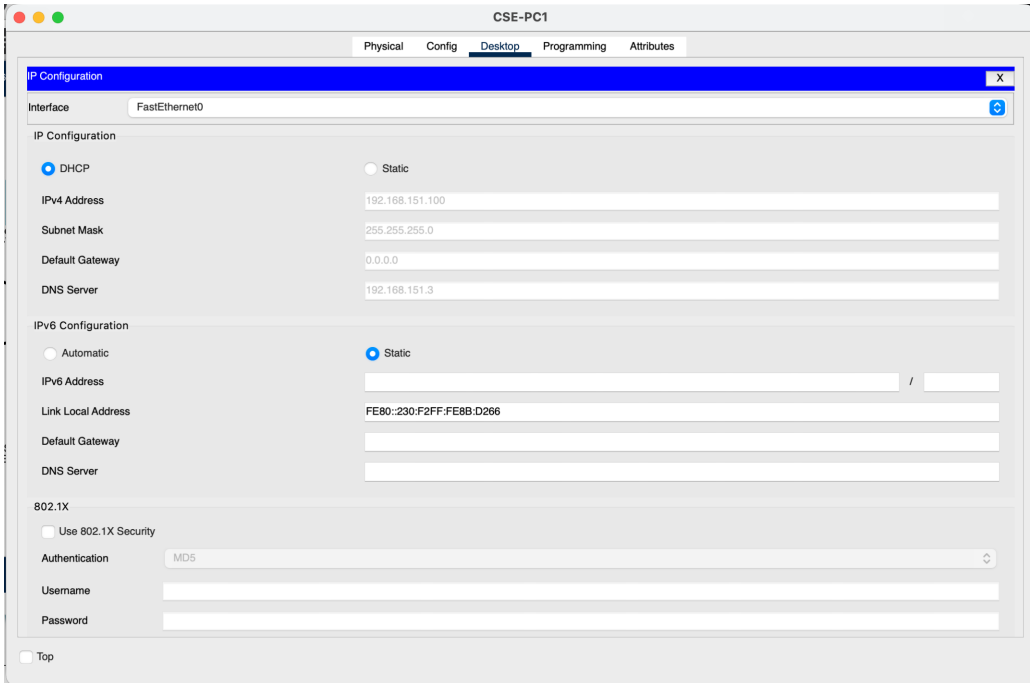


Figure 23: Question 7: Workstation CSE-PC1 IP configuration showing dynamic IP and DNS server (192.168.151.3).

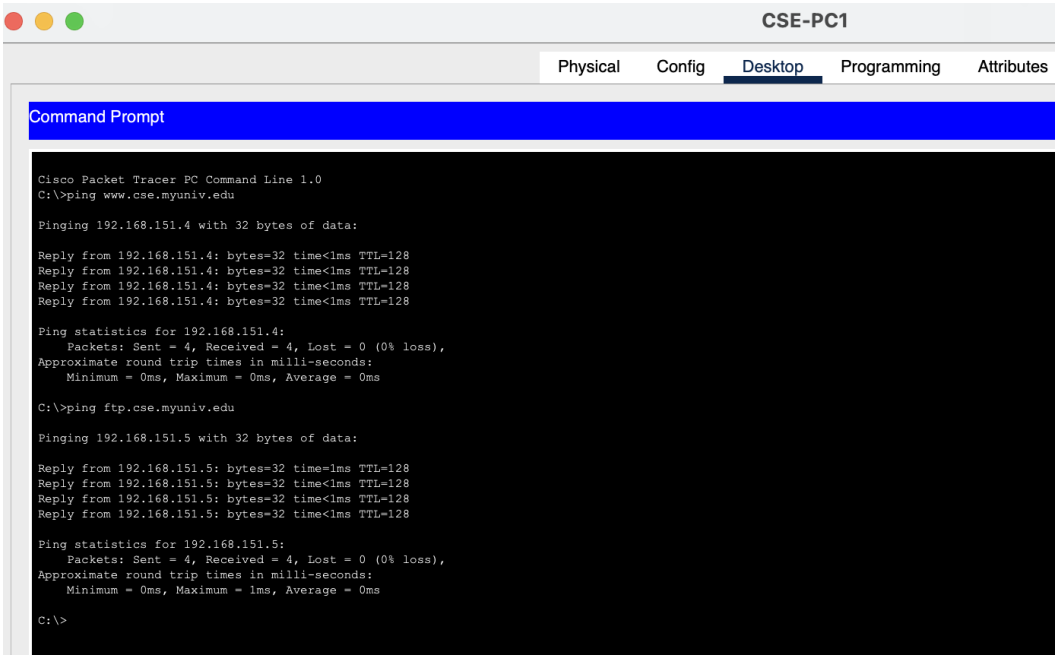


Figure 24: Question 7: DNS resolution verification from CSE-PC1 targeting www and ftp domain names.

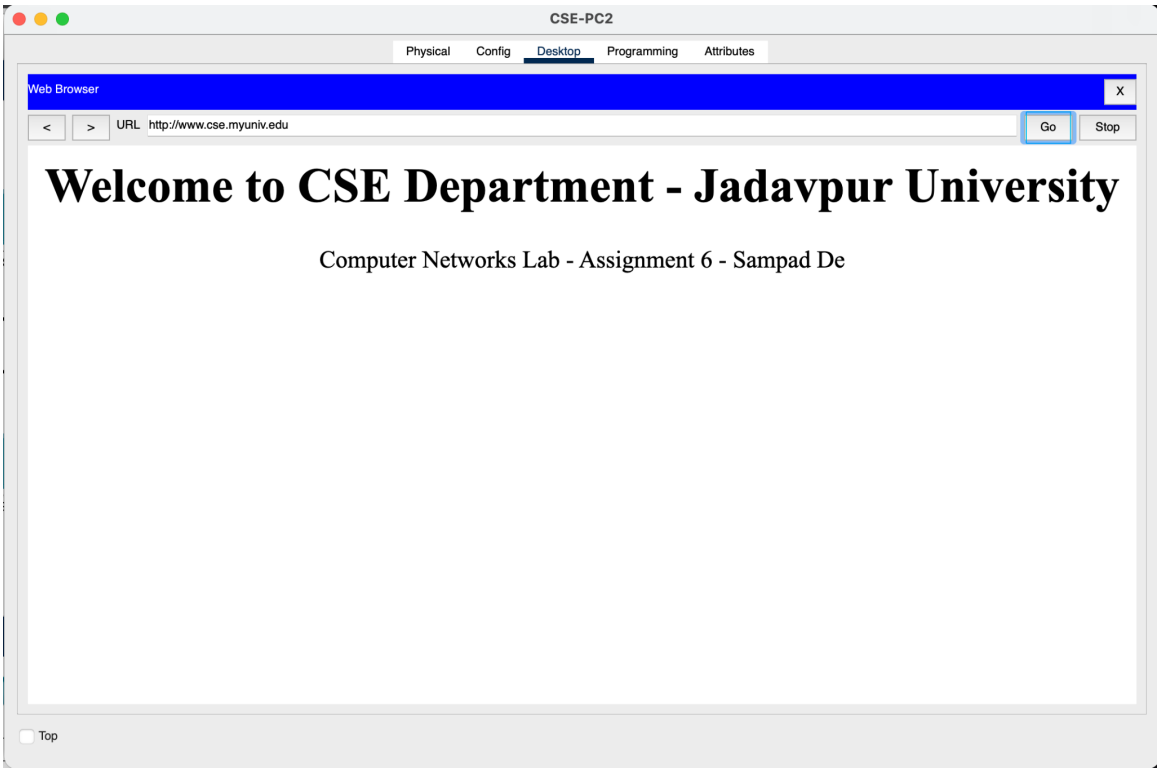


Figure 25: Question 7: CSE-PC2 web browser rendering custom HTTP landing page for Sampad De.

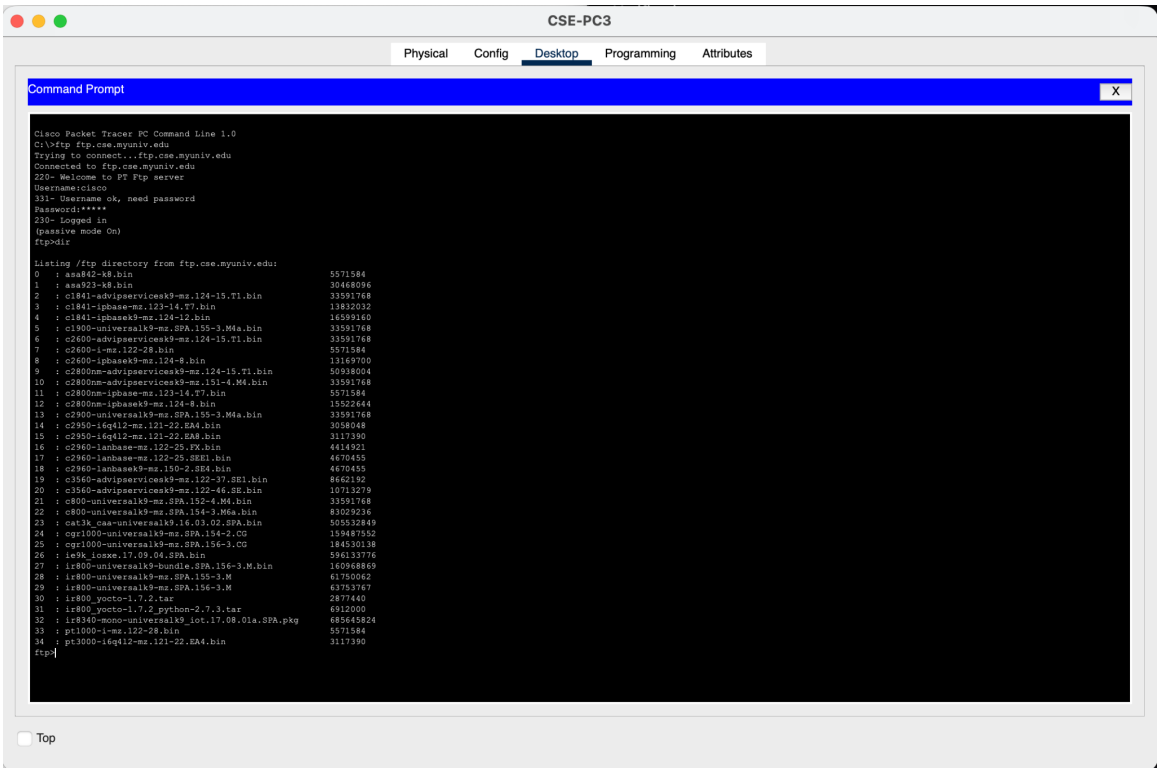


Figure 26: Question 7: CSE-PC3 terminal verifying interactive authenticated FTP session and directory listing.

9 Key Takeaways & Conclusion

1. **Physical Media:** Direct host-to-host links (MDI to MDI) require crossover cabling to pair transmit (TX) to receive (RX) circuitry.
2. **Collision Domains:** Hubs blindly repeat bits across a single collision domain, whereas switches micro-segment ports and learn source MAC addresses into CAM tables to enable full-duplex, selective forwarding.
3. **ARP Mechanics:** Hosts dynamically map 32-bit IPv4 addresses to 48-bit MAC addresses, initially starting with empty tables and populating upon frame transmission.
4. **Subnet Boundaries:** Physical Layer-2 bridging across disparate IP subnets fails without a Layer-3 default gateway to route packets between network IDs.
5. **WAN Routing:** Static routing tables (‘ip route‘) over synchronous serial WAN links enable inter-network routing, accompanied by IPv4 TTL decrementation.
6. **Multi-Tier Services:** Automated IP leasing via DHCP (DORA) and hierarchical name resolution via DNS provide the foundation for robust application-layer delivery (HTTP web and FTP storage).

10 References

1. **Project Source Code Repository:** https://github.com/Sampad2006/CN-Lab-Assignments.
2. **AI Assistance :** AI (Gemini 3.8 Flash) was utilized to generate latex chart configurations and also to use Cisco Packet Tracer. All specific setups,screenshots and answers I have done myself .